

YUEZHEN DONG

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RELEVANT SKILLS

Languages	Python, Java, TypeScript/JavaScript, C/C++, SQL, Ruby, Bash, Go
ML/AI	PyTorch, scikit-learn, Gemini API, RAG, Embeddings, LLM Agents, GRPO, NeMo-RL
Frameworks	FastAPI, Flask, React, JavaFX, Node/Express, Ruby on Rails, Spring Boot, GraphQL
Infrastructure	Docker, PostgreSQL, AWS, Git, Linux, CI/CD, monitoring, distributed training

EDUCATION

University of Waterloo – Faculty of Engineering	Expected May 2028
BASc in Computer Engineering	Waterloo, ON

WORK EXPERIENCE

Meta Engineering Fellow	Jun 2026 – Sep 2026
Python, Bash, Linux, Docker, CI/CD, Networking, Monitoring, Automated Testing	Meta + MLH Fellowship

- Engineered **Python/Bash** automation and Dockerized test harnesses for Linux services, integrating deployment validation, service-health checks, and structured log analysis into CI/CD workflows.
- Built observability and diagnostic tooling that correlated logs, system metrics, and network events to detect resource saturation, configuration drift, and dependency failures, accelerating root-cause analysis across distributed services.

Software Engineering Intern	May 2026 – Aug 2026
Python, PyTorch, TensorFlow, TypeScript, React Native, C++, Ruby on Rails, LLM, GRPO	Shopify

- Raised Sidekick Analytics subagent unit-test pass rate from a **43.4% baseline to 87%** by building an end-to-end agentic RL pipeline with **GRPO/NeMo-RL**, converting production-failure tests into verifiable rewards and scoring grouped rollouts across held-out tests, tool errors, and judge metrics.
- Implemented **RAG-based context retrieval** with embedding reranking and prompt orchestration pipelines for agentic AI tooling infrastructure, reducing irrelevant context fed to the LLM by **~40%** and improving AI-generated code acceptance rate in developer review.
- Built **Quick Device Activation** for Shopify POS using **BLE** pairing, QR/manual-code fallback, and **OAuth Device Authorization** – reducing merchant device setup time from **~5 minutes to under 30 seconds**.

Software Engineering Intern	Sep 2025 – Dec 2025
React, TypeScript, Ruby on Rails, GraphQL	Shopify

- Owned a **staff assignment modal** in Retail Admin end-to-end – built React components and designed **GraphQL** queries with server-side pagination, reducing staff search query response time by **~60%**.
- Shipped full-stack features across **6 systems** – Admin Web, POS Mobile, Shopify Core, Business Platform, Cloud Sync Streamer, and POS Channel – maintaining backward-compatible data contracts and achieving **zero regression**.
- Coordinated with engineering, product, and design to clarify requirements and document tradeoffs, reducing mid-sprint scope changes and contributing to on-time delivery of **3 consecutive merchant-facing feature releases**.

Web Developer Co-op	Jan 2025 – Apr 2025
Java, Spring Boot, Maven, Apache POI, Git	AGF Investments

- Built a **Java/Spring Boot** ScoreCard application with XLSX streaming via **Apache POI**, configurable transformation pipelines, and input validation, automating report generation across **4 departments** and saving **20+ staff hours/week** previously spent on manual spreadsheet processing.

PERSONAL PROJECTS

Multimodal Agent Evaluation Sandbox
Python, PyTorch, Gemini API, FastAPI, React, PostgreSQL, Docker

- Built a **multimodal benchmark** for AI agents evaluating text, image, code, and tabular reasoning through tool-calling, RAG, sandboxed Python execution, and verifier-based scoring – agents evaluated on this benchmark achieved **23% higher accuracy** on unseen task variants compared to a zero-shot baseline.

Ecoland: AI Ecosystem Simulator
Java 17, JavaFX, Gradle, OOP, Neural Networks, Data Logging

- Engineered a **Java 17/JavaFX** agent-based simulator with Perlin terrain, **7 species**, neural-network brains, interactive controls, real-time charts, CSV logging, and save/load serialization.